

Homework 5

COMP221 Spring 2026 - Suhas Arehalli

Complete the problem below. Check the course website & syllabus for further instructions.

If any problem is unclear, or you think you found a typo, please let me know ASAP so I can clarify!

Problems

1. **Making Change** Suppose you are a cashier making change for a customer. Consider the following problem description for the problem of providing change with the fewest number of coins:

PROBLEM (Minimal Change Set).

Input: $k \in \mathbb{Z}$, a number of cents, and $C = \{c_1, \dots, c_n\} \subseteq \mathbb{Z}_+$.

Output: A multi-set (set that allows duplicates) L such that

- $\forall c \in L, c \in C$
- $\sum_{c \in L} c = k$
- $|L|$ is minimized.

if such a set exists. *NULL* otherwise.

That is, I need you to find the fewest possible coins such that the values of those coins add up to k .

You might have seen in Discrete Math that a greedy strategy — adding the largest coin that doesn't go over k — works for US coin denominations (i.e., the case where $C = \{1, 5, 10, 25\}$). In fact, you can prove this correct using a clever, but tedious exchange argument.

- (a) Prove via counterexample that the greedy strategy will not work for all sets of coin denominations C . That is, construct C such that the greedy approach fails to find the smallest multi-set of coins that sum to some k .
- (b) Describe the solution to this problem in terms of a recurrence relation. For simplicity, write a solution that finds the **size** of the optimal set rather than constructing the set itself.

****HINT**:** Think carefully about how to frame the problem as a sequential decision process. It may be helpful to think about a way to avoid the mistake a greedy algorithm makes on your counterexample in the prior part, or to consider the subset sum problem.

- (c) Provide pseudocode for a dynamic programming algorithm that computes the **minimum number of coins** needed to make k units of change that works for *any* coin system. For full points, provide a solution that runs in $O(k|C|)$ time and $O(k)$ space. A solution with a worse space complexity can still get the majority of points.